DesignGrad: Conceptual, Numerical and Visual Feedback for a Local CAD Coding Agent

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Abstract. Mechanical design is an iterative process. This paper studies which kinds of textual feedback improve the designs of a local language model that writes parts as CAD code. **DesignGrad** is inspired by TextGrad: a critic reviews the current design, and the review is converted into instructions for revising it. Gemma 4 26B-A4B runs offline on a laptop and performs every role in the loop. Three kinds of feedback are added cumulatively: conceptual feedback, in which the critic reads the design document; numerical feedback, in which the agent writes and runs its own analysis code to measure the built part and to calculate quantities such as mass, stress and deflection; and visual feedback, in which a reviewer inspects rendered views of the built part. After the loop has finished, independent tests score each design against its requirements. On three design problems with three repeats each, the score increases with every level of feedback, by 28% in total, and the largest increase comes from numerical feedback. The three kinds of feedback also detect different faults: conceptual feedback identifies details that the design document leaves undefined, numerical feedback identifies the requirements that the built part does not meet, and visual feedback often identifies the cause, for example a plate built in the wrong orientation.

1 Introduction

Mechanical design is an iterative process. Engineers begin with a brief problem statement, develop it into a detailed design, model the part and evaluate the result. Each evaluation leads to a revision, and the cycle continues until the part meets its requirements.

As AI moves towards the physical world, agents that can do mechanical design are becoming popular [1–5]. Feedback-driven improvement is well established in software engineering: coding agents perform better when they execute their code, read the errors and test the output [6–9]. Design agents have adopted the same principle, using rendered images of the part [3, 4], geometric measurements from the CAD kernel [5] and finite-element analysis in the loop [10, 11]. However, the contribution of each kind of feedback has not been measured separately. Consequently, little evidence guides the design of a harness for design agents.

DesignGrad is inspired by TextGrad [12], a method for the automatic improvement of prompts and outputs. In TextGrad, a critic evaluates an output, and the evaluation is propagated backwards as a "textual gradient": an instruction for revising the text that produced the output. DesignGrad applies this method to two texts, a design document and a CAD program.

The feedback levels are inspired by how engineers work: they iterate on their design. They receive a problem statement with little detail and develop it into a design document. They review this document to verify that the concept satisfies every requirement. They model the part and analyse the model with measurements and simulations. Finally, they inspect a rendering of the finished part to confirm that it matches their intent. We give a local agent these three evaluations as conceptual, numerical and visual feedback, and ask: **how do its designs improve as conceptual, numerical and visual feedback are added, and how should a local model be used for design?**

The system runs offline on a laptop with one open model, Gemma 4 26B-A4B [13]. For simple parts, local operation makes design agents accessible and inexpensive, keeps design data on the user's machine, and allows deployment on the local computers that are connected to manufacturing equipment. The designs are scored after the loop has finished, by independent tests that are not part of the feedback.

2 The system

2.1 The loop

DesignGrad is a TextGrad-inspired loop [12] that iteratively improves two texts: a **design document**, which states the concept, the dimensions and how each requirement is satisfied, and the **CadQuery program** that builds the part. One Gemma 4 model performs every role in the loop (Fig. 1). The designer reads the specification and writes both texts. The CAD agent executes the program and builds the part. The critic evaluates the design against each requirement and returns a verdict per requirement together with a list of specific problems, each stating the feature, its current value and the required value (Fig. 11). A gradient step converts the critique into revision instructions for the design document and for the program, and the designer rewrites both. The loop runs for at most four rounds.

At the end of each round, the new design is compared with the best design so far, using the critic's verdicts and the agent's own measurements. The next round continues from the new design if it is better, and otherwise restarts from the best design.

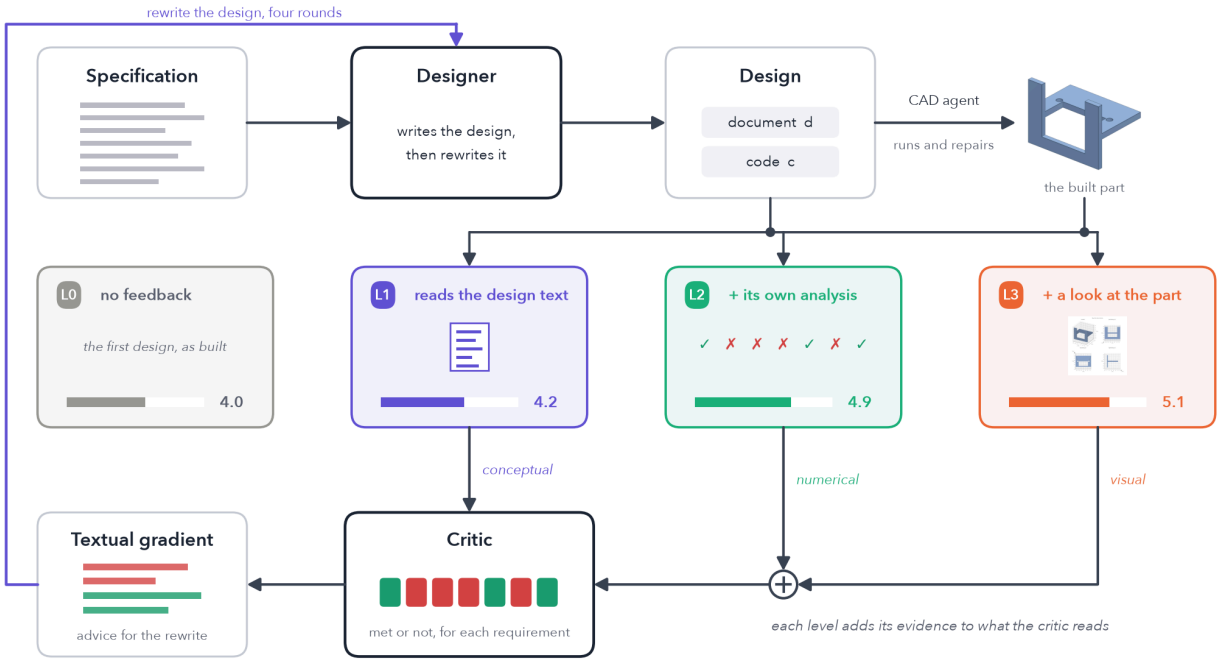


Fig. 1. The DesignGrad loop: design, build, review, textual gradient, revision. Bars: requirements met (of 7) at each level.

2.2 Feedback levels

The four levels differ only in the information that the critic receives. They are cumulative: each level retains the feedback of the levels below, and all start from the same first design.

L0, no feedback. The designer writes the design document and the CadQuery program once, and the CAD agent builds the part. No critic is involved and the design is not revised.

L1, conceptual feedback. The critic receives only the design document and does not see the built part. For each requirement, the critic recalculates masses, stresses and positions from the dimensions stated in the document, and reports every incorrect statement and every undefined dimension.

L2, numerical feedback. The critic additionally receives measurements of the built part. An analysis agent writes a Python program for each problem from the specification. The prompt is identical for all problems and contains only general reference material (library calls and textbook formulas). The program loads the built part and evaluates every requirement. It probes the geometry (whether material is present at a point or a region is empty), measures dimensions and mass, and, for the shelf bracket, applies beam theory to the cross-section at the wall to obtain bending stress and deflection. Because a self-written check can be incorrect, each analysis program is also

executed on a control part, a one-millimetre cube without features (Fig. 2). A check that reports a hole or a wall on the cube does not measure the intended feature and must be rewritten; a measurement that fails this control is not accepted as evidence.

L3, visual feedback. The critic additionally receives a visual review. A reviewer inspects four rendered views of the built part together with the design document and lists every difference between the part and the document.

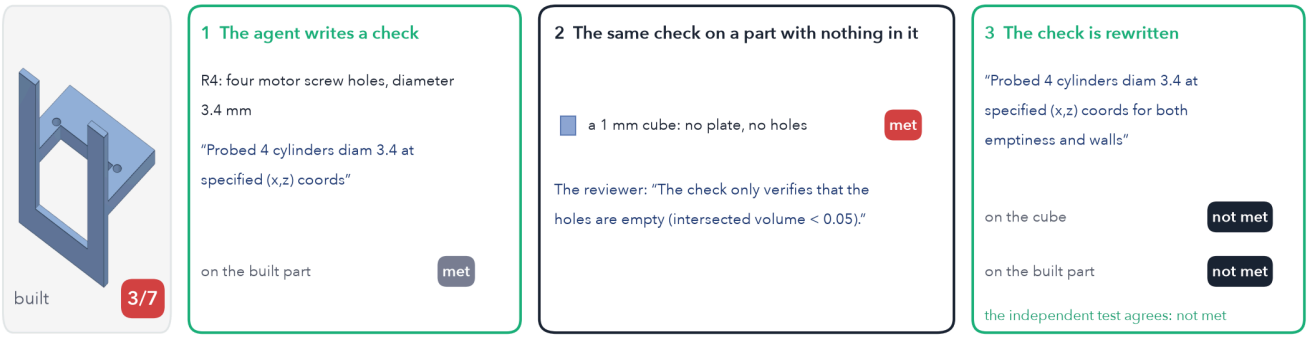


Fig. 2. The control part at L2 (motor mount). The first check for the motor screw holes reports them on a part whose face plate is missing, and also on the empty cube; the rewritten check is correct.

3 Experiment

Problems. The main experiment uses three parts, each with seven numerical requirements (Fig. 4): a shelf bracket that must carry a load, an electronics enclosure base, and a motor mount. Each part is run three times with different sampling seeds, giving nine runs per level.

Independent tests. For each problem, a set of geometric and analytic tests, one per requirement, was written separately from the agents' tools and fixed before any run. A reference design passes all tests and a plain block fails most of them. They are applied after a run has finished. "Requirements met" always denotes the number of these tests that a design passes.

Model. `gemma4:26b` in Ollama on an M1 Max laptop with 32 GB of memory, temperature 0.3.

4 Results

4.1 Results by level

Table 1. Final designs, nine runs per level. Best design produced: the best design of any round.

Level	Requirements met (of 7)	Best design produced	Model calls	Minutes
L0 no feedback	4.0	4.0	2	1
L1 conceptual	4.2	5.0	14	4
L2 + numerical	4.9	5.1	39	14
L3 + visual	**5.1**	**5.7**	44	15

Each level of feedback improves the design (Table 1, Fig. 3). Conceptual feedback alone has a small effect. Numerical feedback produces the largest single increase, 0.7 requirements, and visual feedback adds a further 0.2. From L0 to L3 the designs improve by 1.1 requirements, or 28% (95% interval +0.4 to +1.9); five of nine runs improve and none deteriorates. Each single step lies within the variation between runs, whereas the overall improvement does not.

Within the loop (Fig. 3, top right), the design improves steadily at L2 and L3, whereas at L1 it is unstable, because a critic that cannot observe the built part sometimes prefers a revision whose part is worse.

The three parts respond differently (Fig. 3, bottom; Fig. 4). The enclosure base does not change at any level: one first design already meets all seven requirements, and the other two fail only the screw-boss requirement. The

agent's own check reports the missing pilot holes, but no revision corrects them, so the points in Fig. 3 coincide. The shelf bracket improves when numerical feedback is added (3.3 requirements at L0 and L1, 4.7 at L2, 4.3 at L3); the mass limit, the envelope and screwdriver access most often remain unmet. The motor mount improves at every level, most with visual feedback (2.3, 3.0, 3.7, 4.7).

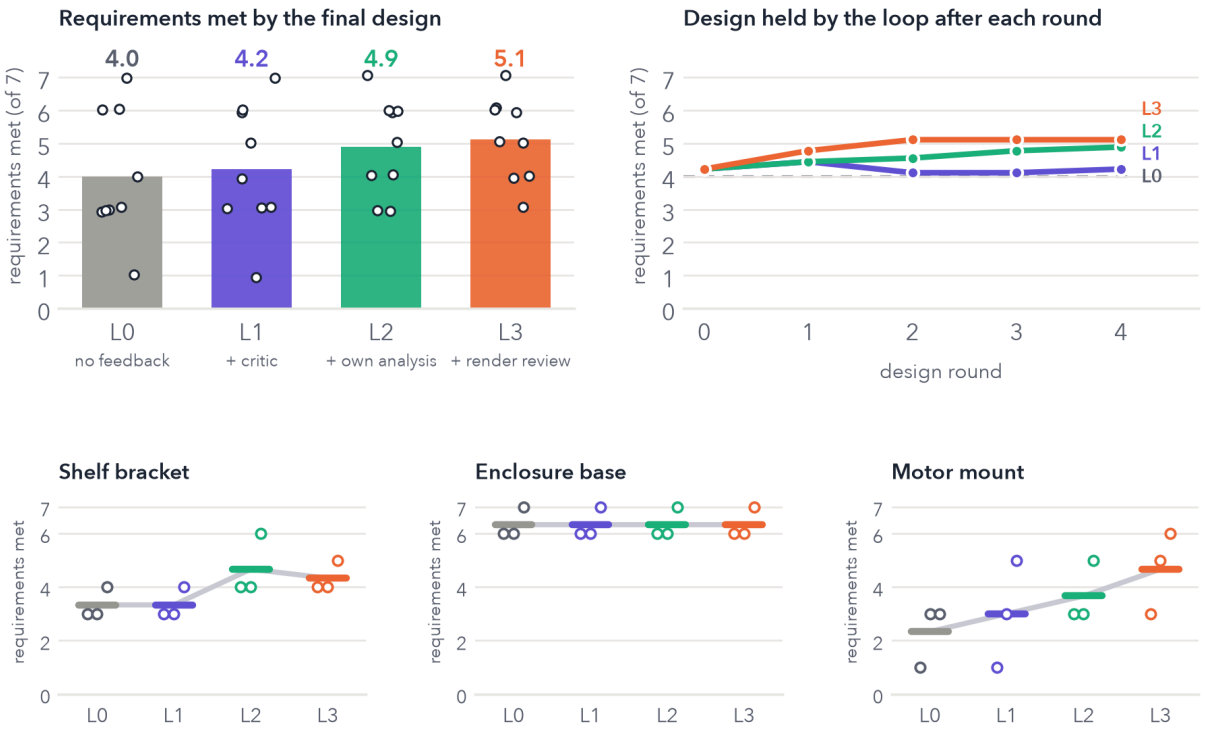


Fig. 3. Top left: requirements met by the final design; each point is one run. Top right: the design held by the loop after each round, mean of nine runs (dashed: L0; the curves start higher because one first design was repaired further). Bottom: each part; points are repeats, bars are means.

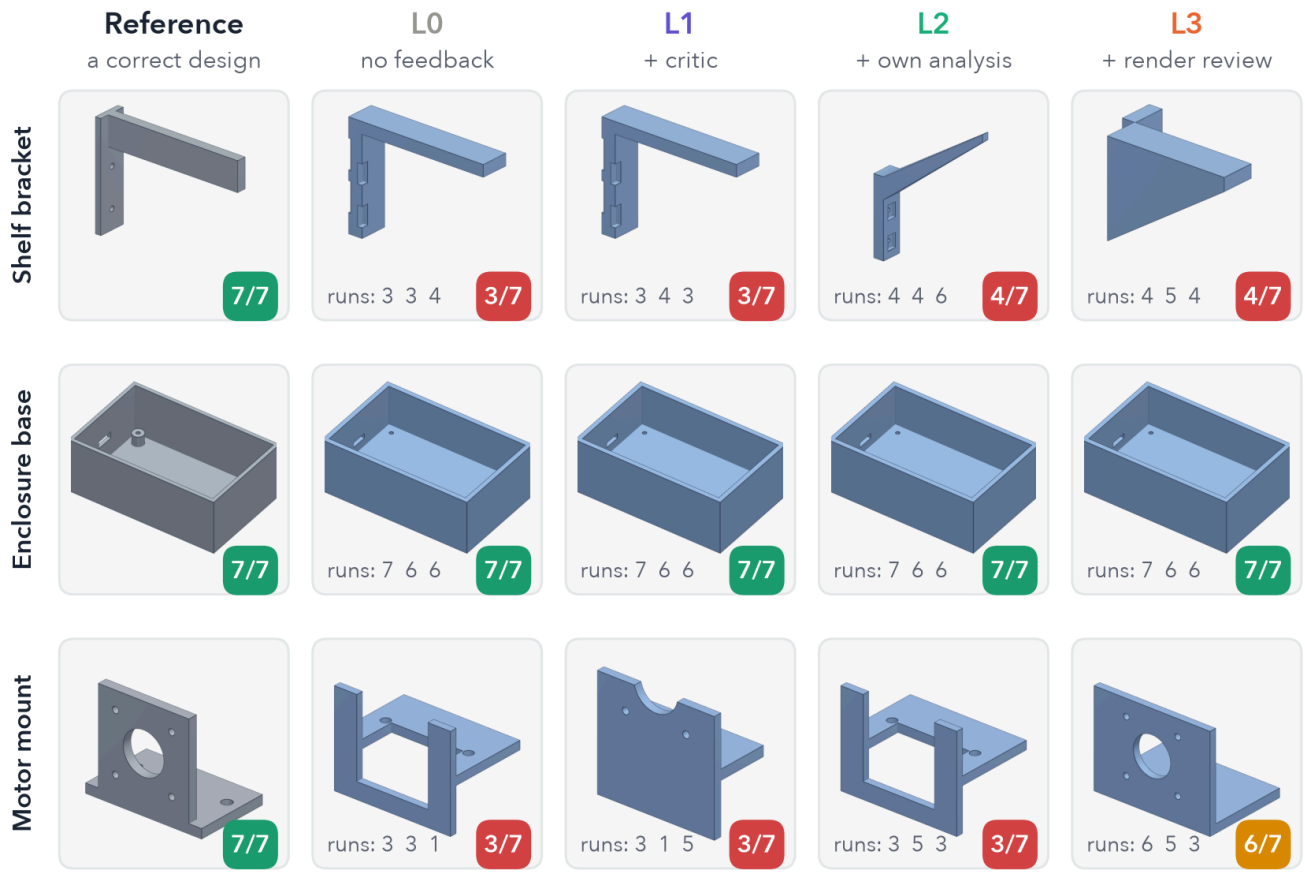


Fig. 4. A correct reference design and the final design of each level, first repeat; "runs" lists all three repeats. **Shelf bracket:** two wall-screw holes with screwdriver access, a seat 110 mm up and 150 mm out, 150 N within stress and deflection limits, at most 90 g. **Enclosure base:** an 85 × 55 mm tray, four screw bosses, a USB-C cutout, at most 70 g. **Motor mount:** a 4 mm face plate with a bore and four screw holes, motor clearance, a base with two bolt holes, at most 60 g.

4.2 Three critics, one design

Fig. 5 shows the three critics evaluating the same first design: a motor mount whose face plate was built in the wrong plane and whose base lies below the table.



Fig. 5. The same first design at each level: the part before, what the critic read and wrote, the change in the design document (red: removed text; green: added text), and the part after. Frame colours: purple L1, green L2, orange L3. Quotes in all figures are verbatim.

At L1 the critic cannot detect this fault, because the design document describes the face plate correctly. The critic instead identifies the undefined dimensions of the clearance pocket, and these are added. The document improves, but the built part does not.

At L2 the analysis program finds neither the bore nor the screw holes at their specified positions, and the critic reports this. The report is correct but describes the symptoms and not the cause. The designer replaces the plate with a solid block, the part remains incorrect, and four rounds pass without progress.

At L3 the reviewer inspects the rendered views and identifies both faults directly. After one revision the part meets six requirements. Numerical feedback indicated what was wrong; visual feedback indicated why.

4.3 Round by round

When a measurement identifies the fault, numerical feedback is effective (Fig. 6, top; Fig. 7). The first shelf bracket weighs 445 g against a limit of 90 g; the designer replaces the solid section at the wall with a hollow section, and the next part weighs 43 g and meets six requirements. The motor mount (Fig. 6, bottom; Fig. 8) improves similarly, one fault per round.

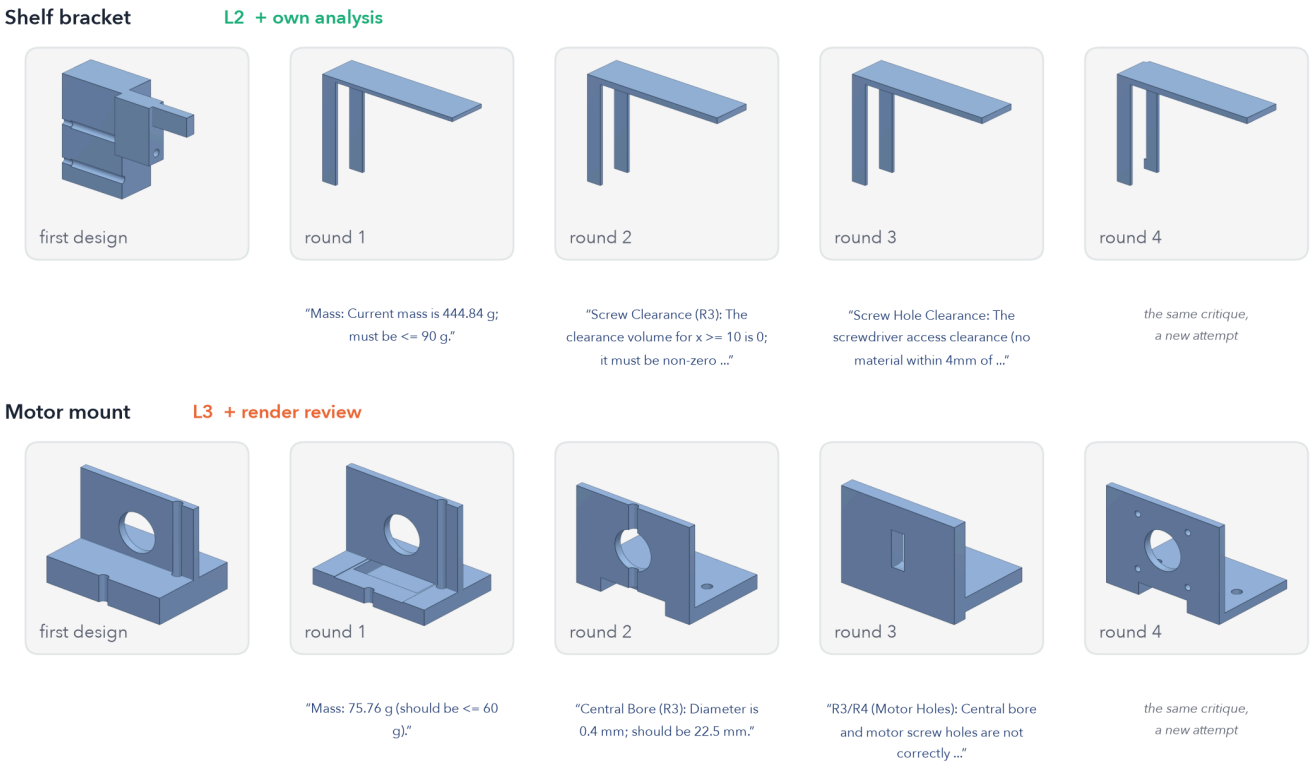


Fig. 6. Two loops, round by round: the part produced in each round and the first problem that the critic named in the preceding design.

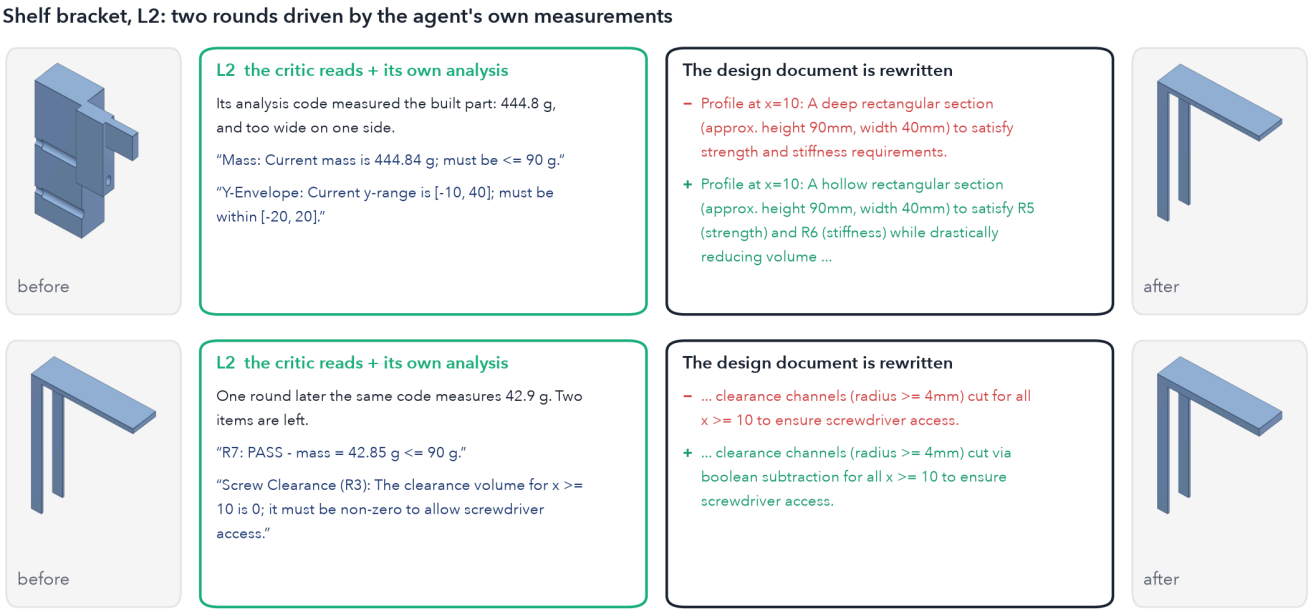


Fig. 7. The first two rounds of the shelf bracket (colours as in Fig. 5).

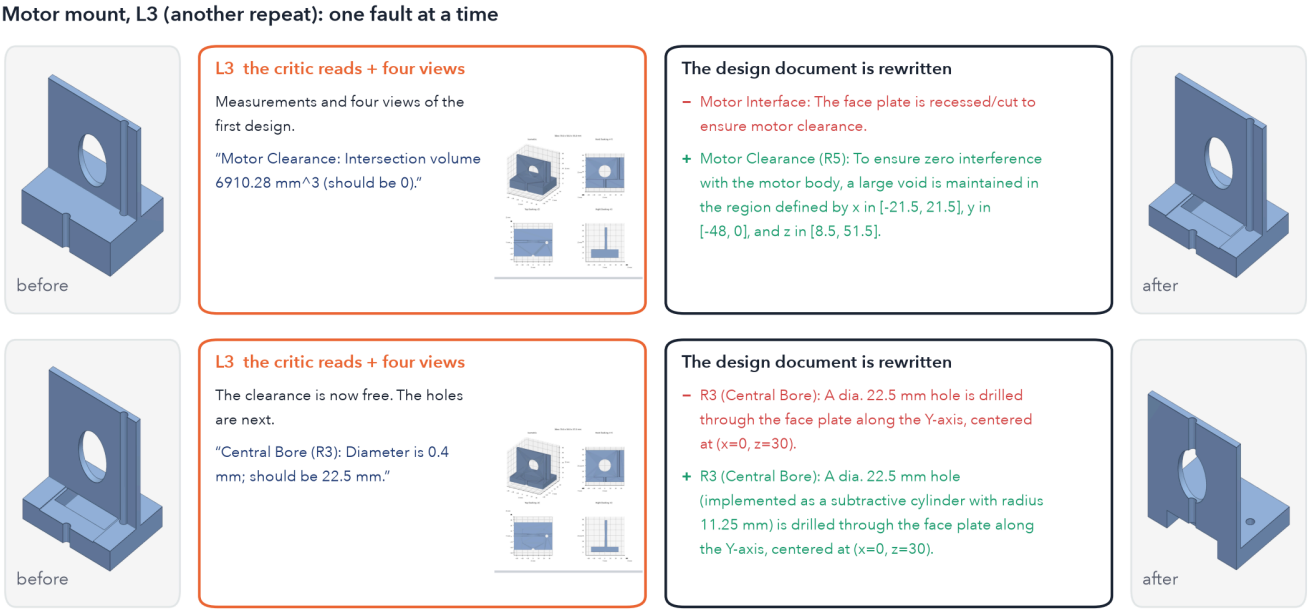


Fig. 8. The first two rounds of the motor mount (colours as in Fig. 5).

4.4 What limits the loop

Two factors limit the loop (Fig. 9). First, detecting a fault is easier than correcting it: a design round corrects only a minority of the unmet requirements and occasionally breaks a requirement that was already met. The correction rate is highest at L3, where the visual review often identifies the cause of the fault. Second, the agent's own evaluation is not always correct. It rarely misses a real fault, but it sometimes reports a requirement as unmet although the part meets it, for example when a check examines a region larger than the feature. Such false alarms can cause the loop to discard a design that is in fact better. The visual reviewer shows the same tendency, which is why visual feedback is most useful together with measurements.

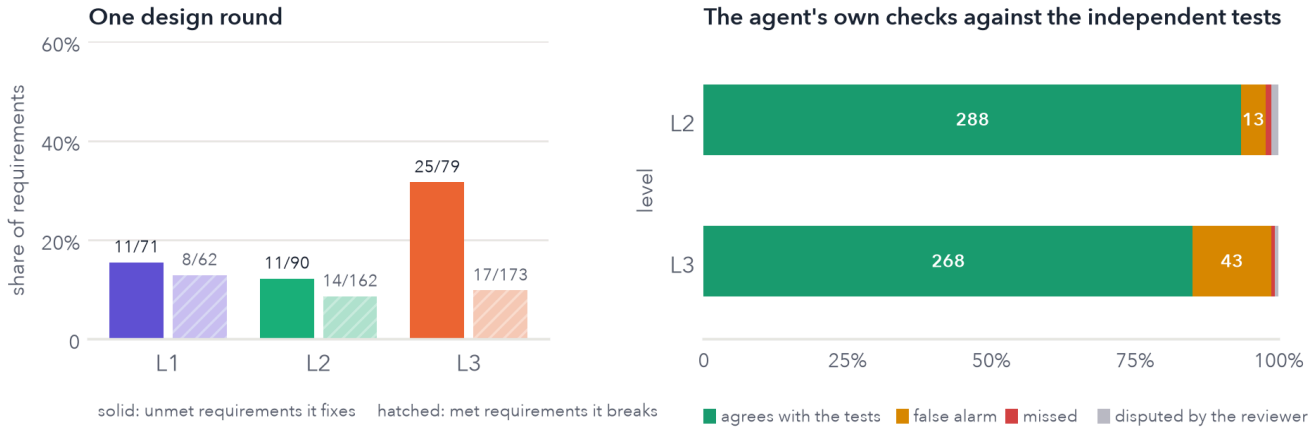


Fig. 9. Left: the share of unmet requirements that one design round corrects (solid) and of met requirements that it breaks (hatched). Right: the agent's own verdicts compared, after the runs, with the independent tests: agreement (green), false alarm (orange), missed fault (red), disputed by the reviewer (grey).

4.5 Other parts

Four further parts were run twice each (Fig. 10): a phone stand, a pipe clamp, a drone arm and a heat sink (six requirements). None was used when the loop was revised (Section 7). The trend is similar: the designs meet 3.9 requirements at L0, 3.6 at L1, 4.6 at L2 and 4.7 at L3; three runs improve from L0 to L3 and none deteriorates. Two runs are excluded because the CAD agent did not complete most of their builds, so their scores do not measure the feedback: a tilted sensor mount (12 of 16 builds; 0, 2, 2, 2 from L0 to L3) and the second pipe clamp run (10 of 16; 2, 4, 3, 5).



Fig. 10. All seven parts: a correct reference, requirements met at each level (mean of the repeats), and the final part at L3.

5 Discussion

The best designs are obtained when all three kinds of feedback are used together: the designs at L3 meet 28% more requirements than the designs without feedback. The largest improvement occurs when numerical feedback is added, because the agent then evaluates the built part instead of its description. Visual feedback provides a further improvement, mainly by revealing the cause of a fault that the measurements only detect. This improvement has a cost: the time per design increases from about one minute without feedback to about fifteen minutes with all three kinds. For a local model, the results therefore suggest that the agent should build and measure every design, verify its own measurements on a control part, and inspect the rendered part. The remaining difficulty is to correct a detected fault without breaking another requirement. Further improvement is expected from the revision step and from testing the part in a physics simulation.

6 Related work

Self-feedback, code execution and self-written tests improve model outputs [6–9], although self-critique without external evidence is unreliable [14]. CAD agents refine their code from rendered views [3, 4], kernel measurements [5] and finite-element analysis [10, 11]. DesignGrad differs in three respects: the model writes its own analysis code, this code is verified on a control part, and feedback is added cumulatively so that each contribution can be measured.

7 Limitations

Claude wrote the problems and the independent tests and assisted with the code; fixing the tests before the runs and validating them on reference and incorrect designs reduces the risk of bias but does not remove it. The

sample is small: nine runs per level, with large variation between runs, and two runs per further part. The loop was revised after first passes over the three main parts, so these parts were not unseen; all reported runs use the final version. The analysis is geometric and analytic, not finite-element.

8 Reproducibility

```
python -m designgrad.run --ladder v6 --problems P01 --levels L0 L1 L2 L3 --out runs/x
python eval/v4_analyze.py runs/x --out runs/x
```

Every model call is recorded; replaying the records reproduces the runs we checked byte for byte. Code, problems, tests and records: https://github.com/sinansahincandan/designgrad



Fig. 11. Four of the prompts, verbatim.

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